

MTH212M Quiz Part B (February 6, 2026)

6:30 pm - 7:45 pm (+25 minutes for DAP students)

Instructions:

1. Write your answers in the designated place. If your answer is not legible, you shall not get credit.
2. If necessary, you may write your answer using a simple fraction.
3. Provide all details/justification including calculations.

Name:

Roll No.:

Question 1. (2 marks) Let $\{X_n\}_{n=0}^{\infty}$ be a Markov Chain on the state space $S = \{1, 2, 3\}$ and with the T.P. matrix $P = (p_{ij})_{i,j \in S}$ given by

$$P = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{6} & \frac{1}{2} \end{pmatrix}.$$

Given that $\mathbb{P}(X_0 = i) = \frac{1}{3}, \forall i \in S$. Compute $\mathbb{P}(X_0 = i | X_1 = 2)$ for all $i \in S$.

Question 2. Prove the following statements.

- (a) (2 marks) Let C be any communicating class. Then no proper subset of C is closed.
- (b) (3 marks) Suppose i is transient and $j \rightarrow i$. Then $\sum_{n=1}^{\infty} P_{ij}^n < \infty$.
- (c) (3 marks) A Markov Chain on a finite state space has at least one recurrent state.
- (d) (2 marks) Consider a Markov chain $\{X_n\}_{n \geq 0}$ on the state space $\{0, 1, 2, 3\}$ and the T.P. matrix

$$\begin{pmatrix} \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 \\ \frac{1}{6} & \frac{1}{2} & 0 & 0 \\ \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 \\ 0 & \frac{1}{6} & \frac{1}{3} & \frac{1}{2} \end{pmatrix}.$$

Then the state 3 is transient.